Plant Biotechnology Lab. BT 421

Lab (5)

Micropropagation and Callogenesis

By:

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Outline

- Micropropagation
- Micropropagation stages
- Multishoot induction
- Root induction
- Callogenesis

Micropropagation

- **Clonal propagation in vitro is called micropropagation.**

Derivation of thousands of plants from a single cell in a short span of time.

- **It is the multiplication of genetically identical individuals by asexual reproduction.**
- **While clone is a plant population derived from a single individual by asexual reproduction.**

Micropropagation stages

1. Initiation stage:

- A piece of plant tissue (called an explant) is (a) cut from the plant, (b) disinfested (removal of surface contaminants), and (c) placed on a medium.

- The objective of this stage is to achieve an aseptic culture.

2. Multiplication stage.

- A growing explant can be induced to produce vegetative shoots by including a cytokinin in the medium.

- A cytokinin is a plant growth regulator that promotes shoot formation from growing plant cells.

3. Shoot transfer to a rooting medium.

- Growing shoots can be induced to produce adventitious roots by including an auxin in the medium.

- Auxins are plant growth regulators that promote root formation.

4. Acclimatization (Hardening).

- A growing, rooted shoot can be removed from tissue culture and placed in soil.

- When this is done, the humidity must be gradually reduced over time because tissue-cultured plants are extremely susceptible to wilting.

1. Shoot Multiplication

A. Multiplication through Callus culture (Indirect).

Growth hormones

- **Drawback: Genetic variation for mass cells.**
- **Less preferred.**

a. Callus formation

Differentiated explant $\xrightarrow{\text{growth factor + 2-4-D}}$ undifferentiated explant (meristematic tissue with genetic variation)

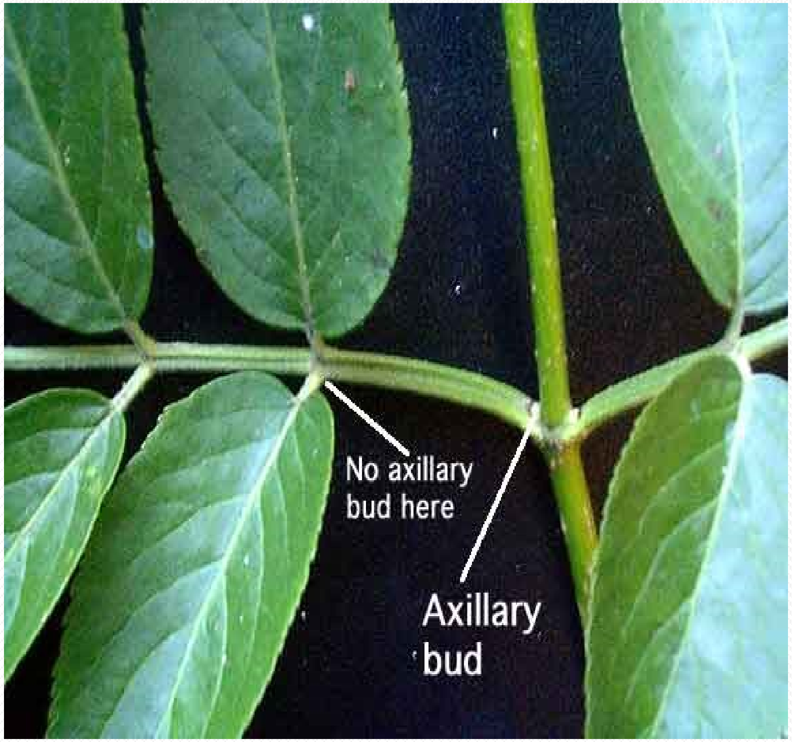
b. Organogenesis

Un differentiated explant with growth factor produce nodule (unfertilized ova) then cuts for nodule and put it inside the media then produce shoot or root .

* Most effect way to produce callus
* Some shoot are not stable

B. Multiplication by Apical and axillary shoots

- No genetic variation



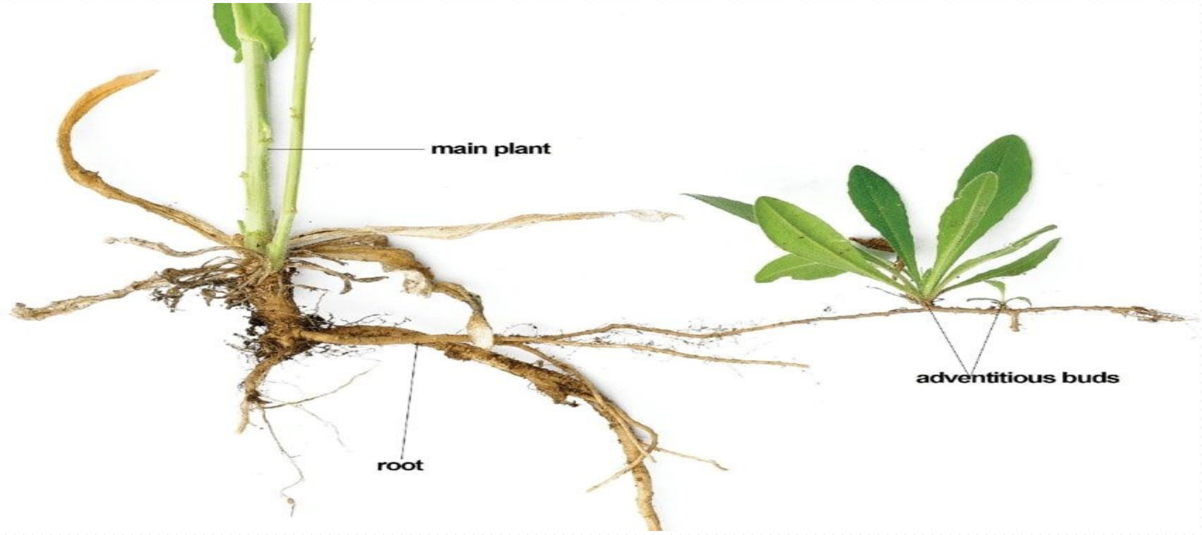
- Don’t pass through callus (no genetic variation)
- Shoots are identical genetically
- Cut axillary bud from it and then put the bud in Cytokine media then give 1-3 shoots

C. Multiplication by adventitious shoots

- Commercial practice

Use combination of growth factors

Cytkinin + GA



2. Root induction

-Initiation of rooting is often the most labour-intensive stage of in vitro propagation.

-All cytokinins inhibit rooting.

-We use Auxin (IBA, IAA) + less salts concentrations + (0.5-1.0% sucrose) in rooting media.

Callogenesis: Callus formation

Callus: undifferentiated mass of cells.

Explant $\rightarrow$ **2-4 D**

$\downarrow$

Auxin=Cyrokinin

Callus aims:

1. **Crop improvement**

We use 0.5 mg/L 2-4 D to avoid toxicity and because it is a mutagen.

2. Multiplication.

-Multishoot induction (shoot proliferation)

By this process make a mutation then have 14% Genetic makeup and 86% Normal

3. Selection.

-Transfer callus to media with different kinds of stresses.

Callus aims:

3. Selection.

-Transfer callus to media with different kinds of stresses.

Make variation in genetic material

Explant → Growth factor 2-4-D → **Callus** → Transfer to → **media with low amount of toxic then to media with medium amount of toxic and finally media with high amount of toxic after that select toxic resistant colony . .**

Callus induction

-Can be initiated from explant taken from any part of plant.

-There are two types of callus:

1. Friable.

   loosely attached give somatic embryo.

   Important in prepare cell culture not tissue culture

   Need for growth liquid form of media

2. Compact.

Callus induction

-It is induced from:

1. Single cell.
2. Multiple cells(10^4- 10^5).

Callus growth patterns